Project II — AI Financial Advisor: Agentic Portfolio Construction

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Business Specifications

The traditional financial advisory industry suffers from structural problems: commission-driven recommendations, AUM fees that incentivize complexity, and allocation templates that ignore a client's most important variable — **their career**. Human capital (present value of future earnings) is typically a person's largest asset, yet portfolio construction ignores it. A tech executive with $2M in RSUs needs a fundamentally different portfolio than a biology professor with a pension. This project builds a multi-agent AI system that is transparent, auditable, and conflict-free to construct personalized portfolios — and critically evaluates whether the AI's recommendations serve the client's interest.

Problem Statement

Can a multi-agent LLM system construct personalized investment portfolios that account for human capital exposure and career-specific risk — and can students critically evaluate whether the AI's recommendations are genuinely in the client's interest?

Data Availability

- **WRDS (CRSP, Compustat)**: Historical returns and fundamentals (Fordham subscription)
- **FRED** (Free): Yield curves, credit spreads, unemployment, CPI
- **LLM APIs**: Claude/ChatGPT for agent implementation ($5–20/student)
- **Three client personas** (provided): Biology professor, tech executive, financial professional
- **Compute**: All analysis runs on a laptop

What Students Build: A Five-Agent System

Agent	Role	Key Outputs
Profile	Structured intake — career type, income stability, industry exposure, holdings, horizon, risk tolerance	Total wealth assessment (human + financial capital)
Research	Scans macro conditions (FRED) and matches against historical regime analogues	Regime assessment with historical comparisons
Allocation	Constructs personalized portfolio with per-position rationale, accounting for human capital	Proposed allocation with justification
Risk	Adversarial — stress-tests recommendations against client-specific risk profile	Concentration, correlation, regime vulnerability
Compliance	Checks fiduciary standards, suitability rules, concentration limits	Clearance or flagged violations

What Students Do

Phase 1: System Design & Data Infrastructure (Weeks 1–4)

1. **Design agent architecture.** Define roles, communication protocols, validation gates.
2. **Build data pipeline.** WRDS for historical returns, FRED for macro. Define 4–5 regimes (dot-com, GFC, COVID, rate hikes, AI boom).
3. **Implement Profile Agent.** Structured intake capturing total wealth (financial + human capital).

Phase 2: Portfolio Construction & Backtesting (Weeks 5–8)

4. **Run system for three personas.** Must produce meaningfully different allocations for each.
5. **Historical regime backtesting.** Backtest personalized allocations through each regime vs. 80/20 benchmark. Key question: when does personalization matter most?
6. **Regime change detection.** System proposes rebalances when macro shifts. Students evaluate: justified or not?

Phase 3: Critical Evaluation & Presentation (Weeks 9–12)

7. **Compare against industry standard.** Document typical advisor recommendation vs. agent system for each persona.
8. **Address "already priced in" question.** Students grapple with market efficiency.
9. **Write paper and prepare defense.** Key question: "Explain why the tech executive got a different allocation than the professor, and what regime change would flip that."

Deliverables

- Five-agent system in Python (Profile, Research, Allocation, Risk, Compliance)
- Three-persona portfolios with meaningfully different allocations
- Historical regime backtesting (4–5 regimes) vs. 80/20 benchmark
- Regime change detection and rebalancing evaluation
- Agent vs. industry comparison + trust evaluation (hallucination detection, fiduciary methodology)
- Academic paper (12–15 pages), oral presentation + Q&A defense, code repository

> **The Trust Question:** The distinguishing feature is not portfolio construction — it is the validation pipeline. How do you know the AI's recommendation serves the client? How do you catch hallucinated alpha? What does "fiduciary duty" mean when the advisor is a machine?

Structure

Duration	12 weeks	**Mentor meetings**	Weekly via Zoom
Team size	~6 students	**Faculty advisor**	Prof. Qing Sheng
Infrastructure	~$0 (laptops + Fordham WRDS)	**Confidentiality**	NDA not required

> **AI Policy:** Students may use AI tools (Claude, ChatGPT) for coding and research. Grading measures *demonstrated understanding* through oral defense and targeted questioning.